

# Introduction

- ▶ In 1956, John Milnor shocked the mathematical community.
- ▶ He constructed smooth 7-dimensional manifolds that are *homeomorphic* but not *diffeomorphic* to the standard 7-sphere.
- ▶ Consider the complex equation:

$$z_1^2 + z_2^2 + z_3^2 + z_4^3 + z_5^{6k-1} = 0,$$
$$(z_1, z_2, z_3, z_4, z_5) \in \mathbb{C}^5.$$

- ▶ These are the so-called **exotic spheres**, a discovery that earned Milnor the Fields Medal.

# Exotic Differential Structures on $\mathbb{R}$

- ▶ Two atlases on the same topological line:

$$\mathcal{A} : \varphi_1(x) = x \quad (\text{standard}), \quad \mathcal{B} : \varphi_2(x) = x^3 \quad (\text{cube}).$$

- ▶ Transition maps:

$$\begin{aligned} \varphi_2 \circ \varphi_1^{-1}(x) &= x^3, & (\text{smooth}), \\ \varphi_1 \circ \varphi_2^{-1}(y) &= x^{1/3}, & (\text{not smooth at } 0). \end{aligned}$$

- ▶  $\Rightarrow$  Atlases  $\mathcal{A}, \mathcal{B}$  are **not equivalent**
- ▶  $\Rightarrow$  Different smooth structures.

# Unique up to diffeomorphism

- ▶ Define

$$f : (\mathbb{R}, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B}), \quad f(x) = x^{1/3}.$$

- ▶ In coordinates:

$$\varphi_2 \circ f \circ \varphi_1^{-1}(x) = (x^{1/3})^3 = x$$

$$\varphi_1 \circ f^{-1} \circ \varphi_2^{-1}(x) = (x^3)^{1/3} = x$$

- ▶ So in both  $\mathcal{A}$ - and  $\mathcal{B}$ -coordinates,  $f$  and  $f^{-1}$  look like the identity and are smooth.
- ▶  $\Rightarrow f$  is a **diffeomorphism of manifolds**.
- ▶ Uncountably many atlases can give different smooth structures, but in low dimensions (like  $\mathbb{R}$ ) they are all diffeomorphic. **Unique up to diffeomorphism.**

# The Standard 7-Sphere as a Bundle

- ▶ The usual 7-sphere  $S^7$  can itself be seen as an  $S^3$ -bundle over  $S^4$ .
- ▶ Think of  $S^3$  as the set of unit quaternions:

$$S^3 = \{q \in \mathbb{H} \mid |q| = 1\}.$$

- ▶ Then  $S^7$  can be realized as pairs of quaternions  $(q_1, q_2)$  with unit norm:

$$S^7 = \{(q_1, q_2) \in \mathbb{H}^2 \mid |q_1|^2 + |q_2|^2 = 1\}.$$

- ▶ Projection  $(q_1, q_2) \mapsto [q_1 : q_2]$  onto quaternionic projective space  $\mathbb{H}P^1 \cong S^4$  gives:

$$S^3 \longrightarrow S^7 \longrightarrow S^4.$$

- ▶ This is the **quaternionic Hopf fibration**.

# Bundles over the Four-Sphere

- ▶ Denote by  $\xi$  the rank-4 oriented  $SO(4)$  bundle over  $S^4$ .  $M$  is the unit sphere bundle of  $\xi$ .
- ▶ Consider a fiber bundle with fiber  $S^3$  and base  $S^4$ :

$$S^3 \longrightarrow M \longrightarrow S^4.$$

- ▶ For every  $(h,j)$  we get a candidate total space  $M(h,j)$ .

$$\begin{array}{ccc} M := S(\xi_{h,j}) & \xhookrightarrow{i} & E(\xi_{h,j}) \\ & \searrow \pi_{\text{sphere}} & \downarrow \pi_{\text{vector}} \\ & & S^4 \end{array}$$

# The Gluing Construction

- ▶ Build  $M(h,j)$  by gluing two copies of  $S^3 \times D^4$  along their boundary  $S^3 \times S^3$ .

$$M = M(h,j) = (S^3 \times D^4) \cup_{\xi_{h,j}} (S^3 \times D^4).$$

- ▶ The gluing map is defined using quaternion multiplication:

$$(u, v) \longmapsto (u, u^h v u^j), \quad u, v \in S^3 \subset \mathbb{H}.$$

- ▶ The integers  $h$  and  $j$  tell us how the fibers twist.

# When is $M$ a sphere? I

- ▶ A **Morse function** on a smooth manifold is a smooth function whose critical points are all non-degenerate.
- ▶ Morse functions capture the manifold's topology via the indices of their critical points.
- ▶ **Reeb's Sphere Theorem:** For any closed, connected  $n$ -manifold  $M$ ,

$$\#\text{Crit}(f) = 2 \quad \text{for some Morse } f \implies M \cong S^n.$$

## When is $M$ a sphere? II

- Define a smooth function  $f : M_{h,j} \rightarrow \mathbb{R}$  using the local charts:

$$f(u, v) = \frac{\Re(v)}{\sqrt{1 + \|u\|^2}}, \quad (u, v) \in U_N \times S^3,$$

$$f(u', v') = \frac{\Re(u'(v')^{-1})}{\sqrt{1 + \|u'(v')^{-1}\|^2}}, \quad (u', v') \in U_S \times S^3,$$

$$\text{with } (u', v') = \left( \frac{u}{\|u\|^2}, \frac{u^h v u^j}{\|u\|} \right).$$

- These expressions agree on overlaps  $\iff h + j = -1$ .
- $f$  has exactly two non-degenerate critical points  $\Rightarrow$  by Reeb's theorem,  $M_{h,j} \cong S^7$ .



$$S^3 \rightarrow SO(4)$$

- ▶ Consider the  $S^4$  as  $D_N^4, D_S^4$  glued along  $S^3$ . All data is the transition map  $g : S^3 \rightarrow G$  up to homotopy.

$$\{S^3\text{-bundles over } S^4\} / \cong \longleftrightarrow [S^3, G] = \pi_3(G).$$

- ▶ An oriented rank-4 real bundle  $\xi \rightarrow S^4$  has structure group  $SO(4)$  and its *unit sphere bundle* has fiber  $S^3$ .

# Compute $\pi_3 SO(4)$

- ▶ Define a map  $\varphi$ :

$$\begin{aligned}\varphi : S^3 \times S^3 &\twoheadrightarrow SO(4), \\ \varphi(u, v) : x &\mapsto u x v^{-1}.\end{aligned}$$

- ▶ The map has kernel  $\{\pm(1, 1)\}$
- ▶ So  $\pi_3(SO(4)) \cong \pi_3(S^3 \times S^3) \cong \mathbb{Z} \oplus \mathbb{Z}$ .

# Bundle Sequence

- ▶ Classifying  $S^3$ -bundles over  $S^4$  is equivalent to maps  $f : S^4 \rightarrow BSO(4)$ .
- ▶  $S^3 = \frac{SO(4)}{SO(3)} \implies \frac{S^3}{SO(4)} = \frac{SO(4)/SO(3)}{SO(4)} \simeq \frac{\text{pt}}{SO(3)} = BSO(3)$ .
- ▶  $\pi_4(BSO(4)) \cong \pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z}$   
 $\implies$  the pair  $(h, j)$  classifies the  $S^3$ -bundle.

$$\begin{array}{ccccc} S^3 & \hookrightarrow & M & \longrightarrow & S^4 \\ \downarrow & & \downarrow & & \downarrow \\ S^3 & \hookrightarrow & BSO(3) & \longrightarrow & BSO(4) \end{array}$$

# Difference Map

$$\begin{array}{ccc} \pi_3(S^3) & \xleftarrow{\delta} & \pi_4(S^4) \\ \downarrow \cong & & \downarrow f_* \\ \pi_3(S^3) & \xleftarrow{\delta'} & \pi_4(BSO(4)) \end{array}$$

- ▶ The lower map  $\delta'$  is the **difference map**:  $(h, j) \mapsto h - j$ .
- ▶ Hence  $M_{h,j}$  is a homotopy sphere  $\iff h - j = \pm 1$ .
- ▶ Hence  $f_*$  must send a generator of  $\pi_4(S^4)$  to a pair of integers  $(h, j) \in \pi_4(BSO(4))$  that differ by  $\pm 1$ .

# Bundle Classification Diagram

- Classification of  $S^3$ -bundles over  $S^4$ :

$$\begin{array}{ccc} \pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z} & & \\ \cong \downarrow & \searrow \varphi & \\ \pi_4(BSO(4)) \cong \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{\varphi} & H^4(S^4) \cong \mathbb{Z} \end{array}$$

- A characteristic class  $\kappa$  is a natural transformation

$$\kappa : [S^4, BSO(4)] \longrightarrow H^*(S^4; \mathbb{Z}).$$

- Once you know the homotopy class of  $f$  (i.e. an element of  $\pi_4(BSO(4)) \cong \pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z}$ ), you can compute the characteristic classes in  $H^4(S^4)$ .

# Computation

- ▶ For the bundle  $\xi_{h,j}$ , denote by  $\alpha \in H^4(S^4; \mathbb{Z})$  the generator.
- ▶ The **Euler class** generalizes linking of fibers:

$$e(\xi_{h,j}) = -(h+j)\alpha \in H^4(S^4, \mathbb{Z}).$$

So  $h+j = \pm 1 \Rightarrow M_{h,j}$  is a topological sphere.

- ▶ The **Chern classes** measure the twisting of the bundle:

$$c_1(\xi_{h,j}) = 0, \quad c_2(\xi_{0,-1}) = -\alpha.$$

- ▶ The **Pontryagin class** arises from Chern classes:

$$p_1(\xi_{h,j}) = 2(h-j)\alpha.$$

# Animation Only I

$$\begin{array}{ccc} \pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z} & & \\ \begin{array}{c} h \quad j \end{array} & \searrow \varphi & \\ \cong \downarrow & & \\ \pi_4(BSO(4)) \cong \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{\varphi} & H^4(S^4) \cong \mathbb{Z} \\ \begin{array}{c} h \quad j \end{array} & & \kappa \end{array}$$

## Animation Only II

$$\begin{array}{ccc} \pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z} & & \\ \begin{array}{c} h \quad j \end{array} & \searrow \varphi & \\ \cong \downarrow & & \\ \pi_4(BSO(4)) \cong \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{\varphi} & H^4(S^4) \cong \mathbb{Z} \\ \begin{array}{c} h \quad j \end{array} & & e(\xi) = -(h+j)\alpha. \end{array}$$



## Animation Only III

$$\begin{array}{ccc} \pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z} & & \\ h & j & \\ \cong \downarrow & \searrow \varphi & \\ \pi_4(BSO(4)) \cong \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{\varphi} & H^4(S^4) \cong \mathbb{Z} \\ h & j & p_1(\xi) = 2(h - j)\alpha. \end{array}$$

# Hirzebruch Signature Theorem

- ▶ But with only these invariants is not enough to detect diffeomorphism classes
- ▶ For any smooth, closed, oriented 8-manifold  $C$ :

$$\sigma(C) = \frac{1}{45} \int_C (7p_2(C) - p_1(C) \wedge p_1(C)).$$

- ▶ The signature  $\sigma(C)$  is a **unique invariant of the smooth structure** in dimension 8.

# Milnor's Invariant $\lambda(M)$

- ▶ Let  $M^7$  be a smooth, closed, oriented 7-manifold. Suppose  $M = \partial B$  for some compact 8-manifold  $B$ .
- ▶ Milnor found certain characteristic class invariants of  $M$  restrict the ways  $M$  can appear as the boundary of a smooth 8-manifold.
- ▶ Define

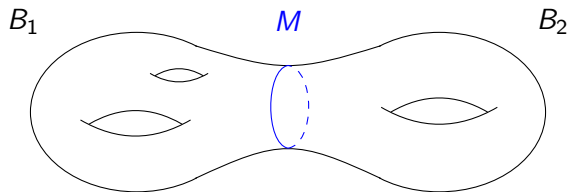
$$q(B) = \int_B p_1(B) \wedge p_1(B), \quad \sigma(B) = \text{signature}(B).$$

- ▶ Milnor's invariant is

$$\lambda(M) := 2q(B) - \sigma(B) \pmod{7}.$$

- ▶  $\lambda(M)$  depends only on  $M$ , not on the choice of  $B$ .

# Illustration I



## Why $\lambda(M)$ is Well-Defined

- ▶ Suppose  $B_1$  and  $B_2$  are two 8-manifolds with  $\partial B_i = M$ .
- ▶ Glue them along  $M$  (reverse orientation on  $B_2$ ):

$$C := B_1 \cup_M (-B_2).$$

- ▶  $C$  is a closed 8-manifold.
- ▶ By the signature theorem:

$$\sigma(C) = \frac{1}{45} \int_C (7p_2(C) - p_1(C) \wedge p_1(C)).$$

- ▶ This forces

$$(2q(B_1) - \sigma(B_1)) \equiv (2q(B_2) - \sigma(B_2)) \pmod{7}.$$

- ▶ Hence  $\lambda(M)$  is independent of the choice of  $B$ .

# Consequence of $\lambda(M)$

- ▶ If  $\lambda(M) = 0$ , then  $M$  could be diffeomorphic to the standard  $S^7$ .
- ▶ If  $\lambda(M) \neq 0$ , then  $M$  is an **exotic 7-sphere**.
- ▶ It's just like the thread on a light bulb socket. **An E27 socket only accepts E27 bulbs**; you can't screw in an E14 or a B22 bulb.

# Modern Approach

- ▶ Milnor's original calculation was based on  $\lambda(M)$ .
- ▶ An alternative approach: we can compute the Pontryagin classes of the desired fiber bundle.
- ▶ Build the *disk bundle* with total space  $N_{h,j} = D(\xi_{h,j})$ ;  $\partial N_{h,j} = M_{h,j}$ .
- ▶ This comes from the same clutching data  $(h,j)$ : locally trivial, so the data on  $M$  *determines*  $N$ .

$$\begin{array}{ccccc} S^3 & \longrightarrow & M_{h,j} := S(\xi_{h,j}) & \xrightarrow{\pi_S} & S^4 \\ \downarrow & & \downarrow i & \nearrow \pi_D & \\ D^4 & \longrightarrow & N_{h,j} := D(\xi_{h,j}) & & \end{array}$$

# From Bundle to Exotic Sphere

- ▶ Cap  $M(h, j)$  with an 8-disk to form  $K_{h, j}$ , an 8-manifold.
- ▶ The extra  $D^8$  cap doesn't add any degree-4 information.
- ▶ So the characteristic class we need on  $K_{h, j}$  is already determined by the original data  $(h, j)$ .
- ▶ Apply Hirzebruch signature theorem:

$$\sigma(K) = \frac{1}{45} \left( 7p_2(K) - p_1(K)^2 \right).$$

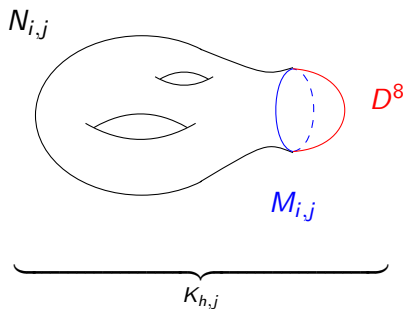
- ▶ Calculation yields condition:

$$(h - j)^2 \equiv 1 \pmod{7}.$$

- ▶ If condition fails,  $M(h, j)$  is an **exotic sphere**.



## Illustration II



- ▶  $K_{h,j} = N_{h,j} \cup_{M_{h,j}} D^8$
- ▶  $H^4(K_{h,j}) \cong H^4(N_{h,j})$

# Visualizing $S^7$

- ▶ In 2012, Niles Johnson gave the first computer visualizations of exotic 7-spheres.
- ▶ His work built on the work of Palais and Cerf: diffeomorphisms and embeddings are tightly linked.
- ▶ For submanifolds  $W \subseteq V \subseteq M$ :

$$\mathrm{Diff}(M) \rightarrow \mathrm{Emb}(V, M) \rightarrow \mathrm{Emb}(W, M)$$

are Serre fibrations (locally trivial).

- ▶ Roughly: if you can wiggle a submanifold, you can extend that wiggle to the whole space.

$$\begin{array}{ccc} \mathrm{Diff}(M) & \longrightarrow & \mathrm{Emb}(V, M) \\ \downarrow & & \downarrow \\ \mathrm{Diff}(W) & \longrightarrow & \mathrm{Emb}(W, M) \end{array}$$

# From Restriction to Visualization

- ▶ Key building block:  $S^3 \subset S^7$ .
- ▶ Restrict diffeomorphisms of  $S^3$  down to embedded circles:

$$S^3 \xrightarrow{\xi_{h,j}^*} \text{Diff}(S^3) \longrightarrow \text{Emb}(S^1 \hookrightarrow S^3)$$

- ▶ This gives a hands-on way to track how  $S^7$  is glued together from circle embeddings inside  $S^3$ .

$$S^1 \xrightarrow{\text{Hopf Fiber}} S^3 \xrightarrow{\xi_{h,j}^*} \text{Emb}(S^1 \hookrightarrow S^3)$$